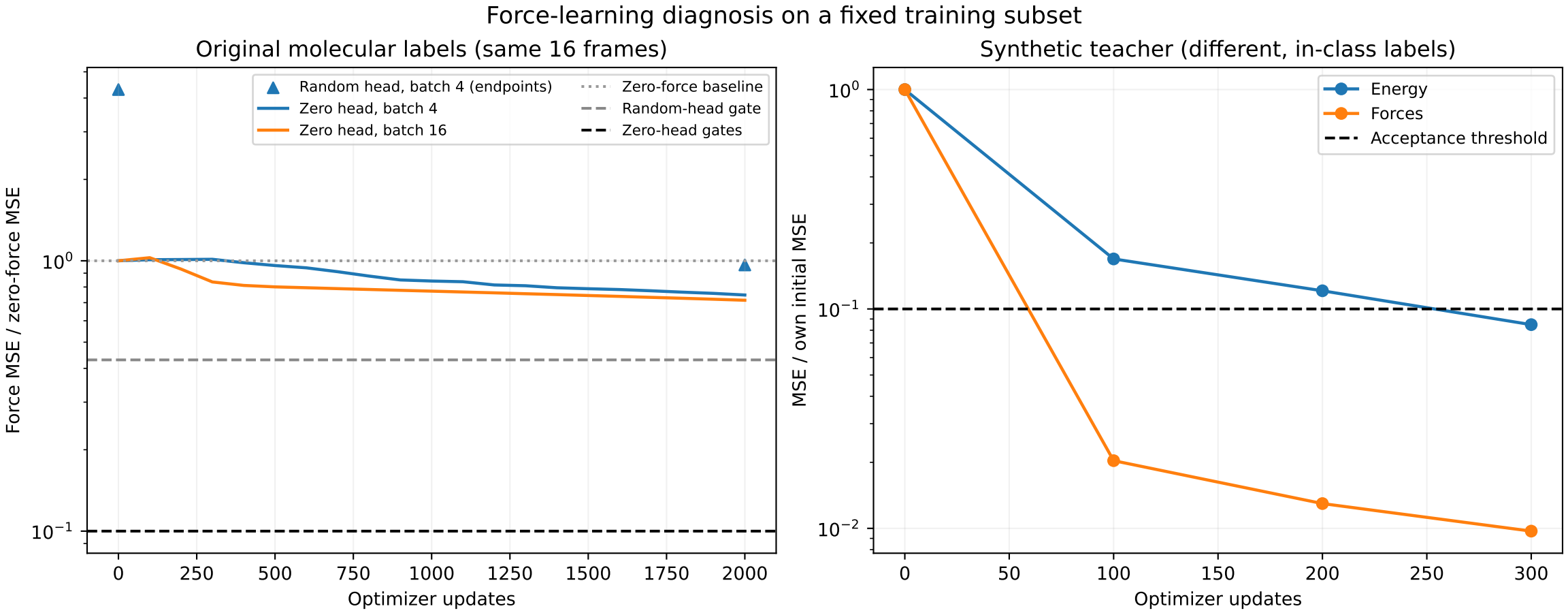
Force-learning diagnosis on a fixed training subset
Original molecular labels (same 16 frames)
Synthetic teacher (different, in-class labels)
Force MSE / zero-force MSE
MSE / own initial MSE
Optimizer updates
Optimizer updates
Random head, batch 4 (endpoints)
Zero head, batch 4
Zero head, batch 16
Zero-force baseline
Random-head gate
Zero-head gates
Energy
Forces
Acceptance threshold
10^{0}
10^{-1}
10^{-2}
0
250
500
750
1000
1250
1500
1750
2000
0
50
100
150
200
250
300